

the gate of a power device through the gate-to-drain or gate-to-collector capacitance. The parasitic turn-on results in lower efficiency or even device failure.

The Miller effect can be alleviated by creating a low impedance path from the power device gate to the source or drain, by increasing the gate resistance, or by driving the gate to a negative voltage with respect to the source or drain. Here, we will discuss the latter option. The goal is to keep the gate voltage below a desired

threshold when a current spike through the Miller capacitance occurs.

For implementing negative drive voltage there is no need to look for gate drivers supporting negative gate drive voltage. One of the common methods is to offset the gate driver from the power device, thereby creating a negative gate drive relative to the source or drain of the power device, while the gate driver IC still sees a unipolar supply only.

The isolated gate driver ICs, however, feature under-voltage lockout (UVLO) to prevent a power device from being driven weakly if the gate driver is being driven with too low of a gate voltage. Two separate voltage sources can be created to tackle this issue. But the trade-off here is with cost.

"Many existing gate drivers that are unipolar in operation can be operated to easily drive a gate negative with minimal external circuitry. There are some implications to consider such as the effective UVLO voltage, but the benefit of such operation is great," says Ryan Schnell, Applications Engineer at Analog Devices.

Moreover, there are certain types of devices that require a negative voltage to be fully off, necessitating some kind of negative voltage drive coming from the gate driver. Fig. 4 shows the

influence of negative drive voltage on switching loss.

3. Controlling switching speed to avoid unexpected turn-on. As discussed, there are three ways to avoid the problem of Miller effect. First, by creating a low impedance path from the power device gate to the source or drain. Second, by increasing the gate resistance. Third, by driving the gate to a negative voltage with respect to the source or drain.

To create a low impedance path from the gate to source, we can add a capacitor (gate-source capacitor). The addition of capacitance produces the same effect as increasing the gate resistance. Both of these will limit the maximum switching frequency by reducing the dV/dt .

To avoid unexpected turn-on, recommended gate-source capacitance is two times value on the specification sheet.

4. Managing dead time loss. Dead time loss occurs in an FET when the current flows from the source to drain while the FET is turned off. In silicon FETs, a body diode is present which typically turns on at about 0.7V. GaN HEMTs do not have a P-type material, hence the body diode is absent.

The conduction from drain to source happens when the drain drops below the gate by more than the threshold voltage of the device. When this happens, the FET turns on backwards with the gate-to-drain voltage turning on the device. The drain falls much below the threshold voltage of the transistor when the FET is on,

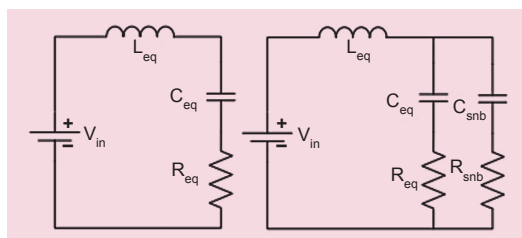


Fig. 7: RLC equivalent model (left) and equivalent circuit with snubber circuit (right)

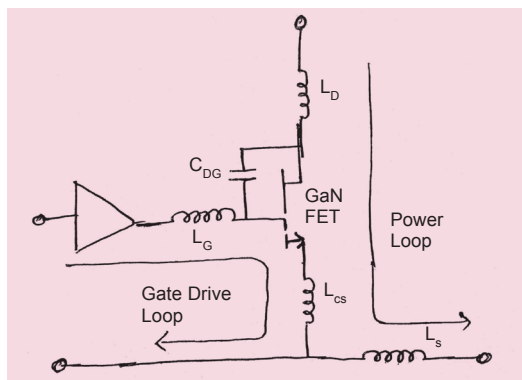


Fig. 8: Circuit with device and parasitic inductances and capacitances

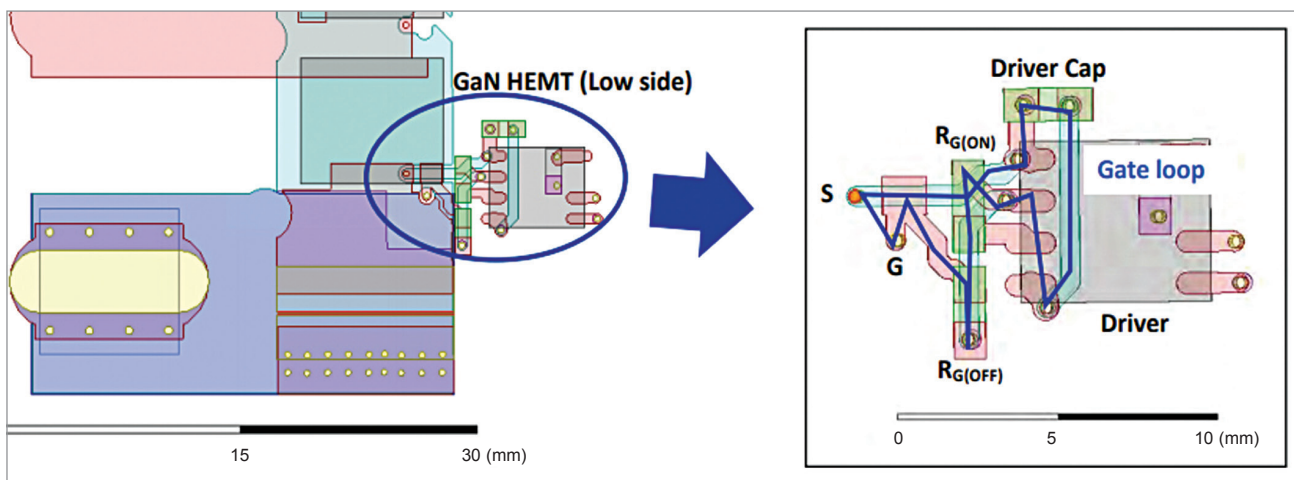


Fig. 9: The gate driver placed close to the GaN FET and minimise the gate drive loop area and length for optimising gate drive loop